Unit-1

1. Explain about the mobile fading characteristics.
2. What is the need of cellular mobile communication systems?
3. Explain about Hexagonal shaped cells.
4. Write any 4 differences between FDD and TDD.
5. Explain uniqueness of mobile radio environment.
6. Explain the operation of a cellular system.
7. Describe the performance criteria of Cellular mobile systems.
8. Discuss about evolution of mobile radio communications.
9. Explain various Analog and Digital Cellular system with an example.
10. Discuss about classification of wireless systems with an example.
11. Explain the various components of a basic cellular system in detail.
12. What are the limitations of conventional mobile telephone system?

Unit-2

1. Why is the cellular concept incorporated?
2. Discuss about channel assignment strategies.
3. What is frequency reuse distance. How it is measured.
4. How to calculate number of channels in a cellular system
5. Define co-channel interference. Discuss about cochannel interference reduction factor.
6. Discuss about frequency reuse concept with diagram
7. Explain about general description of the problem in cellular radio system design.
8. Derive the desired C/I from a worst case in an Omni directional Antenna System.
9. How the interference is different from Apply noise and explain different types of interference in a cellular system. How co-channel interference is measured at the mobile unit and at the cell site?
10. Derive the desired C/I from a normal case Apply in an Omni directional Antenna System.
11. Explain cell splitting and its effect on the performance of cellular systems.
12. What is cell sectorization? Explain briefly?

Unit-3

1. How is Dropped Call Rate calculated? What are the key parameters involved in DCR calculation?
2. Discuss the advantage of delayed handoffs with an example.
3. Explain the differences between hard handoff and soft handoff. How do they impact network performance and mobile device experience?
4. Discuss power difference handoffs.
5. What is meant by Cell site hand off and Intersystem Handoff? Explain.
6. What is Handoff and explain how a handoff is initiated in cellular systems.
7. What is forced handoff? Explain about controlling and creating a handoff.
8. Explain types of handoffs and why they needed?
9. Explain briefly Mobile Assisted Handoff (MAHO) and soft handoff?
10. Discuss about how a two-level hand-off algorithm is useful in delaying a handoff.
11. Discuss about queuing of handoffs.
12. Define the dropped call rate and explain how it is related to the capacity and voice quality.

Unit-4

1. Explain why Multiple Access Techniques are needed.
2. Define capacity in case of TDMA,FDMA,CDMA
3. Explain space division multiple access techniques.
4. What is frequency hopping?
5. What is the near-far problem in CDMA, and how does it impact network performance?
6. Discuss about the features of TDMA.
7. Discuss about the features of CDMA.
8. Compare TDMA, FDMA, SDMA, CDMA.
9. Explain Frequency Division Multiple Access techniques and mention its merits and demerit
10. What are the types of multiple access techniques? Explain detail.
11. Explain Time Division Multiple Access techniques and mention its merits and demerit
12. Discuss about Code Division Multiple Access techniques and mention its merits and demerit

Unit-5

1. Discuss the frame structure for GSM.
2. Write about GSM channels?
3. Compare LTE and WiMAX.
4. How does digital cellular technology improve call quality and data transfer rates compared to analog systems?
5. Explain the principle of WCDMA and write its advantages and disadvantages.
6. List the GSM specifications.
7. What are the key features and benefits of LTE compared to previous wireless technologies?
8. Describe the features and services GSM.
9. Discuss about architecture of WiMAX in detail.
10. Explain the architecture of GSM.
11. What are the key differences between 2G, 3G, 4G, and 5G digital cellular systems?
12. Discuss 5G Technology in detail and mention its advantages and disadvantages.